Optimal minibatch size of SGD-NICE

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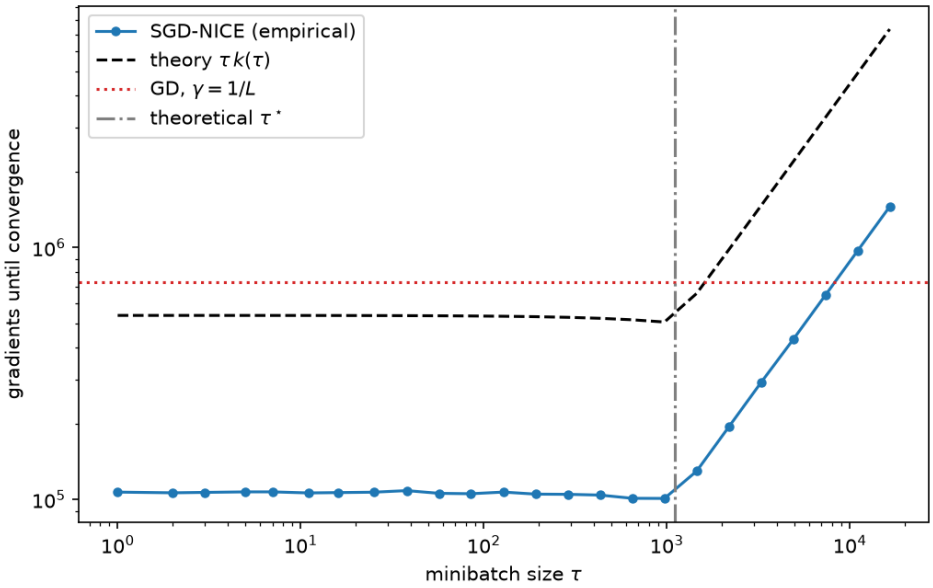
Idea

The idea is to find empirically the best minibatch size τ of SGD-NICE and to compare it with the theoretical optimal minibatch size $\tau^\star$. We use the same linear regression on the California housing data as in `hw_2026_11_11/sgd_lin.ipynb`, with $n = 16512$, so all the constants μ, L, $\max_i L_i$ and $\sigma_\star^2$ are known.

For τ from 1 to n we run SGD-NICE with the stepsize of the convergence theorem and count how many gradients are computed until $\mathbb{E}\left[\left\|w^k - w^\star\right\|^2\right] \le 2\delta \left\|w^0 - w^\star\right\|^2$ with $\delta = 0.005$ (the expectation is estimated with 10 runs). One iteration of SGD-NICE costs τ gradients, and one iteration of GD costs n. The theoretical curve is $\tau\, k(\tau)$, where $k(\tau)$ is the number of iterations from the theorem.

The code is the file `Homework/Exercises/hw_2026_11_20/optimal_batch_size.ipynb`.

Result



The theoretical optimal minibatch size is $\tau^\star \approx 1106$, and the empirical one is $\tau = 971$. Up to $\tau^\star$ the number of gradients is almost constant (about 10^5), and after it the number grows linearly with τ. The empirical curve has the same shape as the theoretical one and is about 5 times below it.